Ontologies, Multiagent Digital Twin and Intelligent Telemetry for Internet Infrastructure (DT-II)

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***Abstract*—This paper presents DT-II, an ontology-driven and multi-agent Digital Twin for Internet Infrastructure. Grounded on the Unified Foundational Ontology (UFO) and modeled in OntoUML, DT-II provides an upper-ontology core centered on the triad `Node--Connection--Channel`, formalized in Description Logic and operationalization as a Knowledge Graph. DT-II continuously synchronizes the ABox with model-driven telemetry (e.g., YANG-Push) and applies automated reasoning over the TBox/RBox/ABox to support semantic interoperability, fault diagnosis, and verifiable governance (e.g., SLAs and Semantic Jurisdiction). The result is a multi-agent, MAPE-K-based architecture that enables explainable, policy-driven actions over heterogeneous network domains.**

***Index Terms*—domain ontology, upper ontology, ufo, ontouml, netconf, multiagent digital twin, internet infrastructure**

I. Introduction

Internet Infrastructure (II) has become too dynamic and heterogeneous to be governed reliably through manual procedures, ad-hoc scripts, and vendor-specific interpretations. At the same time, ***Knowledge Graphs*** (KGs) are increasingly used to represent operational state as connected facts, while ***Ontologies*** provide explicit semantics and constraints for those facts [1]. In DT-II, the ontology defines the logical framework (classes, relations, axioms), and the KG represents its continuously updated instantiation.

This paper proposes an ***Upper Ontology*** (UO) for II [2], grounded on the Unified Foundational Ontology (UFO) and expressed in Description Logic (DL) [3]. The UO functions as a **Logical Constitution** that unifies heterogeneous subdomains (e.g., 5G, satellite, optical) around the triad **Node–Connection–Channel**. OntoUML/OLED is used to separate rigid identity-bearing types (*Kind*) from context-dependent functions (*Role*), enabling systematic model validation and export to OWL/Turtle [4]. An ontology-grounded approach was chosen over purely statistical or rule-based monitoring precisely because it yields explicit, auditable semantics for cross-domain governance and explainability — properties that black-box heuristics cannot provide by construction.

A. Contributions

The main contributions of this work are: (i) an ontology-driven reference model for II centred on *Node–Connection–Channel*, grounded on UFO and engineered in OntoUML; (ii) a DL/OWL operationalization supporting automated reasoning over ***TBox***, ***RBox***, and ***ABox***; (iii) a synchronization approach aligning the KG with model-driven telemetry (e.g., YANG-Push) for near-real-time state updates; (iv) a governance mechanism based on *Semantic Jurisdiction* — a normative layer, defined by ontological axioms and mediated by relators, that determines which constraints (legislation, SLAs, privacy policies) apply to entities and data flows and propagates them across nodes, connections, and channels — enabling verifiable, explainable compliance checks; and (v) a multi-agent Digital Twin architecture following the ***MAPE-K*** cycle [5] for policy-driven actions.

B. Development Road Map

Figure 1 shows the proposed development architecture, which articulates a Proof of Concept (PoC) in Containerlab with a Minimum Viable Product (MVP) spanning three operational complexity profiles: ASN++ (high-complexity, global transit), ASN+ (medium-complexity edge/access), and IX (traffic-exchange domain). While the Upper Ontology (TBox+RBox) is grounded in UFO/gUFO, the Domain Ontology (ABox) is built through continuous PoC–MVP interaction: the PoC offers fast, low-risk iteration over modelling problems in OntoUML/Protégé and Python before facing the assertion volume generated by continuous telemetry (gNMI, YANG-Push) across the three MVP profiles, while the profile diversity gives the academic/IETF community evidence that the ontology unifies domains with distinct operational and legal roles rather than a single network type. The entire ecosystem is maintained and reconfigured in near real-time by the Digital Twin through model-driven telemetry and automated reasoning.

In practice, the PoC links ontology and Python by exporting OntoUML models to OWL/Turtle and ingesting them via RDFLib scripts. Telemetry events are mapped to ontology classes (`Node`, `Connection`, `Channel`), and incremental reasoning is triggered: e.g., a BGP session change instantiates an ABox update and derives `InactiveConnection`. This

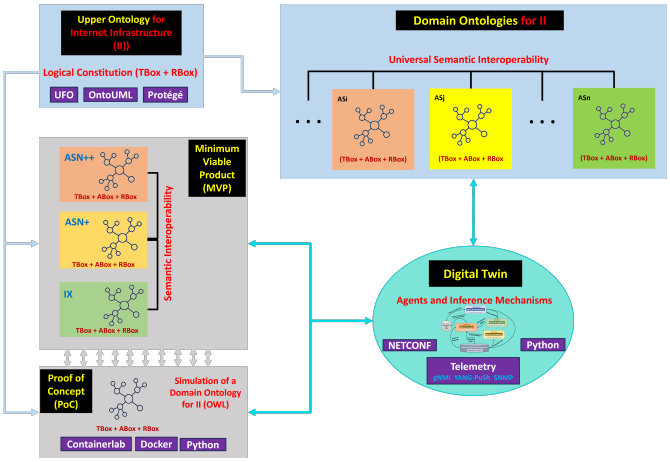


Fig. 1: Architecture of the PoC and MVP proposal across the three operational complexity profiles described in the text, showing the continuous PoC–MVP interaction used to solve domain-ontology abstraction and modeling problems.

loop shows how ontology actively guides code execution in Fig. 1.

On 2026-09-16, the Internet had approximately **79,441** active Autonomous Systems (ASs) [6], as reported in the **CIDR IPv6 Report** maintained by Geoff Huston (APNIC: https://www.cidr-report.org/). Each AS has operational autonomy governed by protocols and specialised administrators, so a misconfiguration in one (e.g., an incorrect BGP announcement) can propagate across the global II. DT-II proposes ontology-based intelligent telemetry that mitigates such risks without human intervention, alongside protocols such as NETCONF, already widely deployed by major II equipment vendors.

The project employs two types of ontologies: domain ontologies (DOs), which specify the knowledge of each AS, and the UO, which ensures semantic interoperability and facilitates the construction and verification of DOs through the OntoUML tool. Although DOs are similar across most ASs, the innovation of the project lies in the use of the UO, which brings benefits such as ensuring semantic interoperability across different domains and facilitating the systematic construction and verification of DOs via OntoUML.

Additionally, the project uses *Protégé*, supported by gUFO, for modeling DOs both in the testbed and in ASs participating in the second phase of the research. The testbed, developed with ***Containerlab***, ***Docker***, and ***Python***, constitutes the DT-II PoC, enabling validation of the project's implementation.

The DT is the operational component of the project. It adopts the autonomic characteristics described in Richard Murch's book "Autonomic Computing" [7], aiming for self-knowledge of its resources and capabilities, autonomous self-configuration and self-optimization, self-healing and self-protection against inconsistencies and threats, contextual awareness in heterogeneous distributed environments, and the ability to anticipate evolving user requirements.

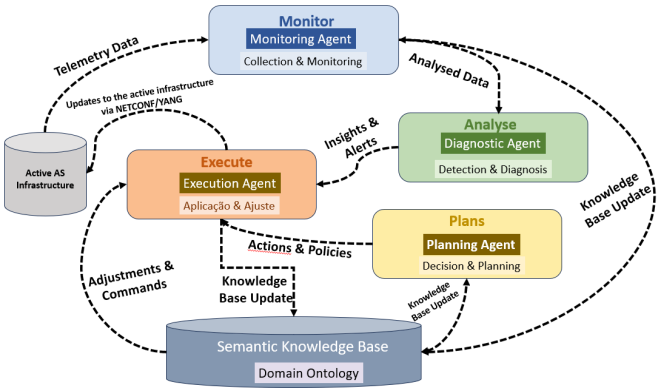


Fig. 2: MAPE-K Flow of the Digital Twin

A DT can preserve these properties if conceived as a **multi-agent** architecture supported by a semantic knowledge base. To this end, the DT follows the **MAPE-K** proposal [5], which establishes the following cycle as a means of ensuring autonomy: **(M) Monitoring** → collects real-time data; **(A) Analysis** → interprets the data based on axioms and rules; **(P) Planning** → generates action plans consistent with the Normative Ontological Basis; **(E) Execution** → applies changes to the infrastructure and **(K) Knowledge** → feeds back into the knowledge base, ensuring continuous learning.

Figure 2 provides a detailed visual representation of the MAPE-K flow of the DT proposed in this article.

II. Related Work

This section positions DT-II with respect to four research lines. **Network Digital Twins and Autonomic Control**: Digital Twins provide a digital counterpart of an operational system, enabling monitoring, diagnosis, and actuation [8]. In networking, recent proposals emphasize closed-loop control, model-driven operations, and integration with telemetry [9]. DT-II differs by making the twin *explicitly governed* by an upper-ontology Logical Constitution, so that actions are validated against semantics and constraints rather than only against metrics or learned policies. **Ontologies and Knowledge Graphs for Network Infrastructure**: Knowledge graphs have been used to integrate heterogeneous network data sources and support cross-domain queries [**?**]. Ontology-driven approaches contribute explicit meaning, identity criteria, and constraints, enabling consistency checking and explainable inferences [1], [3]. DT-II builds on this line by grounding the core model on UFO (via OntoUML/gUFO), explicitly modeling roles and relators, and using this structure as the basis for interoperability between multiple administrative domains [10]. **Telemetry, Model-Driven Interfaces, and Reasoning**: Modern network management increasingly relies on streaming telemetry and model-driven interfaces (e.g., YANG-based mechanisms) to maintain near-real-time state. DT-II adopts model-driven telemetry via YANG-Push (RFC 8641) [11] as a synchronization mechanism, and combines it with DL reasoning over ***TBox/RBox/ABox*** to detect inconsistencies

and derive governance-relevant states (e.g., SLA violations, jurisdiction conflicts). **Governance, Policy, and Compliance**: While policy frameworks typically enforce constraints through configuration templates or procedural checks, DT-II formalises governance as part of the ontology itself (e.g., Semantic Jurisdiction, SLA relators), enabling automated validation, traceability, and explainability of compliance decisions across domains.

Beyond general background, several ontology-based approaches have been proposed in networking. For example, [12] demonstrates the use of ontologies for network management, but focuses on conceptual modeling without addressing dynamic telemetry integration. Similarly, [13] highlights the relevance of ontologies for interoperability, yet provides no implementation details. The ITU-T standards on knowledge management [14] and experimentation engines [15] establish accepted frameworks for closed-loop control and digital twin experimentation, but remain at a high-level specification. In contrast, DT-II contributes a concrete ontology-driven architecture that operationalizes Description Logic reasoning and stream telemetry at ABox level, aiming to bridge the gap between theoretical frameworks and executable governance in Internet Infrastructure.

III. Methodology

This project follows the ontology engineering life cycle [16], using the ***Unified Foundational Ontology*** (UFO) as a basis and OntoUML as the modelling language [17]. The process is structured in four technical phases: M1. **Conceptual Abstraction**: UFO theory is applied to classify network assets. Rigid essence classes (`PhysicalElement`, `MaterialMedium`) define identity, while accidental roles (`NetworkNode`, `CommunicationChannel`) are linked by Relators; M2. **Formalisation in Description Logic (DL)**: Concepts are axiomatized in DL, ensuring computational interpretability. For example, every `NetworkNode` must have a logical identifier: $\mathit{NetworkNode} \sqsubseteq \mathit{PhysicalElement} \sqcap \exists \mathit{possesses}.\mathit{LogicalIdentifier}$; M3. **Semantic Serialisation**: Models validated in ***OLED*** are exported to OWL and serialised in ***Turtle***. Standardised vocabularies such as ***Dublin Core*** (DC) and ***FOAF*** provide support for metadata and provenance; M4. **Automated Reasoning**: Inference engines confront the ***TBox*** and the ***ABox*** to check consistency and detect anomalies. Predicate logic allows deduction of failure states; for example, an inactive Channel implies compromise of its logical connection.

A. Ontological Analysis Structure and Mapping Process

The mapping between the II domain and UFO, via its gUFO implementation, follows a specialization protocol grounded on logical meta properties. For each local domain entity, the **Ontological Analysis Structure** is applied [18]: (a) **Specialisation of Fundamental Types**: Entities with rigid identity and specific individuation principles (for example, Physical Routers, Satellites, Fibre Cables) are mapped as subclasses of *gufo:Kind*. These represent substantial objects that persist independently of their configuration state; (b) **Relational Specialization (Roles)**: Concepts whose existence depends on a specific topological or functional context (for example, Edge Node, Peering Point, Transit AS) are characterized as *gufo:Role*. This allows the Digital Twin to capture the dynamic nature of network functions independently of the underlying hardware; (c) **Relational Foundations**: Logical connections and service agreements that establish dependencies among multiple objects (for example, BGP Sessions, SLA Agreements) are formalized as *gufo:Relator*. This ensures that the integrity of the Knowledge Graph is maintained through the existential dependency constraints inherent to gUFO, where the invalidation of a relator triggers the logical removal of associated roles. **Relators** are UFO entities that objectify relations — e.g., a contract or a BGP session — and are existentially dependent on the entities they connect.

By employing this structure, DT-II transcends mere data modeling, establishing a **Logical Constitution** in which network behaviour is governed by fundamental ontological axioms rather than purely heuristic or statistical patterns.

B. Dynamic Synchronisation via YANG-Push

A critical requirement for DT-II is high-fidelity synchronization between physical infrastructure and its semantic representation. To achieve this, the model employs *Model-Driven Telemetry* (MDT) via the **YANG-Push** mechanism (RFC 8641 [11]).

Unlike traditional query-based (*pull*) monitoring, YANG-Push allows Network Nodes to directly stream state changes to the *Semantic Ingestor*. This integration is mapped into the gUFO framework as follows: (a) **Periodic Subscriptions**: Quantitative metrics, such as interface counters, are streamed to update `gufo:Quality` assertions in the ABox; (b) **Change Notifications**: Critical events, such as oscillations in BGP sessions, are treated as `gufo:Events` that immediately trigger re-evaluation of existential dependencies in `gufo:Relators`.

This ensures that the Digital Twin's Logical Constitution always reflects the current operational state of ASes with sub-second latency.

C. Scalability and Dynamicity at ABox Level

DT-II mitigates scalability by partitioning the ABox into logical slices (e.g., per-AS or domain), enabling localized reasoning and reducing global overhead. Dynamicity is addressed through Stream Reasoning, where incremental engines update only affected triples instead of re-evaluating the full graph. Thus, a BGP session change triggers updates in its partition, while SLA constraints are checked asynchronously. This combination supports near real-time responsiveness at Internet scale. This SROIQ-based strategy trades computational efficiency for expressivity; future work will empirically contrast it against lightweight OWL profiles, such as OWL 2 RL with rule-based reasoners (e.g., RDFox) or OWL 2 QL combined with Ontology-Based Data Access (OBDA), quantifying the resulting trade-off between reasoning completeness and latency under streaming telemetry. Partitioning follows

administrative boundaries (per-AS or per-domain): a BGP session change triggers incremental re-classification confined to its own partition, while cross-partition constraints (e.g., SLA chains spanning multiple ASes) are reconciled asynchronously through scheduled batch checks, bounding worst-case latency to the partition size rather than to the global ***ABox***.

IV. Relations between Ontology, Knowledge Graph and Description Logic

The ***Knowledge Graph*** (KG) operationalises the ontology by linking **real data** (instances) to the conceptual model. It represents information as **nodes** (entities) connected by **edges** (relations). For example, an instance `inst:Router_R1` can be linked to `inst:Router_R2` via `ii:connectedTo`, and the resulting `ii:Connection` can be linked to a `ii:Channel` with capacity and latency qualities.

Ontology and KG are **complementary**: the ontology provides definitions, rules and abstract logic (the "DNA"), while the KG instantiates concrete data, facts and connections that enable queries and discovery (the "organism").

Description Logic (DL) underpins modern ontologies (e.g., OWL on the Semantic Web) [3], enabling formal concept definition and computational reasoning: e.g., $Director \sqsubseteq Person$ states that every Director is a Person, while $Filmmaker \equiv Person \sqcap \exists directed.Film$ — with $\sqcap$ denoting intersection (AND) — defines a Filmmaker as a Person who has directed at least one Film.

In summary, the ontology provides the logical and semantic structure, the KG materializes it in concrete data, and DL formally links the two, ensuring consistency, interpretability and inferencing capability. This triad constitutes the methodological core of DT-II, enabling II to be represented, monitored and governed semantically and dynamically.

V. Conceptual Structure of the Digital Twin

This section presents a **reference architecture** for implementing the Internet Infrastructure Digital Twin (II) as an **operational KG** (***ABox***) governed by a ***UO*** (***TBox***, ***RBox***). The objective is to make the model executable: collect, normalize, validate, infer and support decisions. The implementation objectives are: (a) **Unified semantic inventory**: represent nodes, connections and channels regardless of technology; (b) **Continuous synchronization**: align the KG with telemetry and topology changes; (c) **Detection and diagnosis**: identify inconsistencies, faults and cascading impacts; and (d) **Verifiable governance**: encode and audit rules (SLA, jurisdiction, authorization).

A. Reference Architecture

The architecture is organised into seven layers: (i) **Ontological Layer (*TBox*)**: central classes (`NetworkNode`, `Connection`, `Channel`) and constraints; (ii) **Identity Layer**: stable IRIs separating identity from state; (iii) **Ingestion Layer**: collects and normalizes telemetry (YANG-Push, gNMI, SNMP, syslog) into the Node–Connection–Channel triad; (iv) **Persistence Layer (*ABox*, *RBox*)**: stores facts in RDF/OWL triple stores with temporal history via observations; (v) **Validation Layer**: structural and logical consistency checks with DL reasoners; (vi) **Reasoning Layer**: infers states (`DegradedConnection`, `SLAViolation`) and propagates failures; and (vii) **Exposure Layer**: SPARQL queries, dashboards, alerts and NOC/orchestration integration.

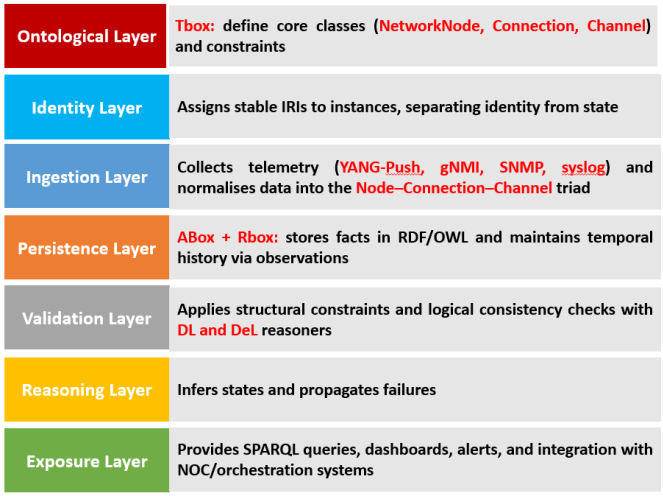


Fig. 3: Digital Twin Reference Architecture

B. Minimum Data Model

The initial implementation focuses on three entities: (a) **Node**: identity (`hasIdentifier`), operational state, timestamp; (b) **Connection**: endpoints (`connectNode`), supported channel, metrics (latency, loss, jitter), and (c) **Channel**: medium type (fibre, copper, radio), capacity, QoS policy.

C. Integration of *ABox* and *RBox*

The ***ABox*** instantiates infrastructure elements (routers, links, channels) from inventory data [19]. The ***RBox***, based on $\mathcal{SROIQ}$ logic [20], governs relational properties: (a) **Transitivity**: deduces cascading failures; (b) **Mereology**: validates part–whole relations for SLA compliance [21, Chap. 8], and (c) **Property chains**: enable bidirectional navigation and anomaly tracing.

Together, ***ABox*** and ***RBox*** turn the KG into a computable model, enabling dynamic reasoning and governance.

D. Operational Flow

The DT-II operational flow unfolds in four concise phases. **F1 Ingestion** collects telemetry (YANG-Push, gNMI, SNMP) and normalizes it into RDF triples instantiating the `Node--Connection--Channel` triad. **F2 Validation** confronts these triples with TBox/RBox axioms to check integrity and detect anomalies (e.g., an active channel without a valid connection), following the partitioning strategy described in Sec. III-C. **F3 Inference** applies Description Logic rules to classify states and derive governance-relevant facts, such as `InactiveConnection` or `SLAViolation`. Finally, **F4 Query/Alert** exposes results via SPARQL queries,

dashboards, or automated orchestration (NETCONF), enabling either operator guidance or direct reconfiguration. Together, these phases transform telemetry into validated knowledge and actionable decisions.

E. Example Scenario

For instance, a fiber link failure is ingested as telemetry, validated as inconsistency, inferred as `InactiveConnection`, and exposed as an SLA violation triggering redundancy.

VI. Use Cases

A. Use Case I: Semantic Jurisdiction and Logical Governance in DT-II

Semantic Jurisdiction is the governance mechanism of the **DT-II** that establishes authority and policy compliance over digital entities based on meaning, context, and relations—transcending traditional geophysical boundaries. In this model, jurisdiction is no longer a static hardware attribute but a **normative entity mediated by Relators**.

Foundation and Axiomatization: Defined in the ***TBox*** through restriction axioms, Semantic Jurisdiction leverages the Reasoner to determine which norms (e.g., LGPD, GDPR, SLAs) apply to real-time data flows. In the ***RBox***, property chains formalize propagation: if a `NetworkNode` is bound to a legal norm, all `Channels` and `Connections` mediated by Relators inherit the same restrictions.

Role of the Digital Twin: The Digital Twin acts as custodian of jurisdiction. By ingesting telemetry (gNMI, YANG-Push, SNMP), the system populates the ***ABox***. Through **Semantic Elevation**, it verifies whether, for example, a health data packet traverses a node under a jurisdiction that fails to meet privacy requirements defined in the ***TBox***.

Operationalization and Resilience: In dynamic events (e.g., DDoS attacks, BGP failures), Semantic Jurisdiction guides the **MAPE-K** cycle [7]: (a) **Analysis**: Reasoner detects violations of jurisdiction axioms; (b) **Planning**: System searches for compliant alternatives; (c) **Execution**: Configuration is applied via ***NETCONF*** only after logical validation, transforming infrastructure into a ***jurisdiction-aware*** system where compliance is verified by DL inference.

B. Use Case II: Behavior of the Model under a DDoS Attack

In a Distributed Denial of Service (DDoS) attack, response time is critical. The Digital Twin acts as a Semantic Immune System, leveraging the triad of ***TBox***, ***RBox***, and ***ABox*** within the ***MAPE-K*** cycle (Fig. 4): (i) **Monitoring** (steps 1–2): telemetry (gNMI, YANG-Push, SNMP) updates the ***ABox*** in real time; e.g., a flow spike from 1 to 100 Gbps is recorded as `inst:Link_01 a ii:Saturated`; (ii) **Analysis** (3–4): the Reasoner applies ***RBox*** property chains: (a) the saturated link supports dependent services; (b) transitivity propagates the risk, inferring `inst:Service_DNS a ii:AtRisk`; (c) the origin is traced to an AS under a specific **Semantic Jurisdiction**; (iii) **Planning** (5–7): consulting the ***TBox***, the system applies rules: if a critical service is at risk and the origin is external, mitigation (blackholing or rate limiting)

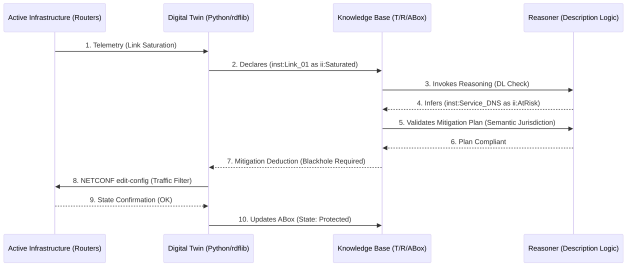


Fig. 4: Sequence Diagram of DDoS Mitigation via DT-II

must occur at the edge, and the Reasoner validates the plan before release; (iv) **Execution** (8–10): via ***NETCONF***, edge routers are reconfigured to filter malicious traffic and the ***ABox*** is updated, closing the loop from the **Active Infrastructure** to the **Logical Core** without human intervention.

Unlike conventional defenses that simply block IPs, this ontology-based approach enables context-aware, policy-driven mitigation: cooperation agreements mapped in the ontology can trigger governance APIs instead of blocking; non-vital channels can be sacrificed to preserve critical services (mereological resilience); each mitigation step is explainable via ***TBox*** axioms; and no action executes without validation against the **Semantic Jurisdiction**.

VII. Conceptual Expansions of DT-II

In this section, we delve into two theoretical aspects that complement the initial proposal of DT-II, complementing Sections III and V and anticipating implementation challenges. **Inter-Domain Multi-agent Coordination**: Although the MAPE-K architecture is already described for internal agents, it is necessary to detail how multiple Digital Twins from different Autonomous Systems (ASes) interact. Inter-agent communication protocols, such as FIPA-ACL or KQML, can be employed for message exchange between graphs. In scenarios of distributed attacks (e.g., DDoS), planning agents from different ASes must negotiate routes and mitigation strategies, establishing consensus or federation among multiple MAPE-K cycles [22]. **Conflict Resolution in Logical Constitution**: Semantic Jurisdiction evaluates plans against strict rules, but conflicts are inevitable in real infrastructures — for example, the self-protection rule may collide with SLA requirements for high availability. To address such situations, we suggest the use of defeasible logic or hierarchies of ontological weights, allowing the inference engine to resolve conflicting axioms in a consistent and transparent manner [23].

VIII. Conclusion and Future Works

Internet Infrastructure has become too complex for purely procedural management: this paper introduced the conceptual foundations of **DT-II**, a multi-agent Digital Twin grounded in Ontology, Knowledge Graphs, and Description Logic, that reframes II as a jurisdiction-aware, autonomic system in which compliance is mathematically verified rather than assumed. By formalizing governance through **Semantic Jurisdiction** and enabling policy-driven responses to complex events such

as DDoS attacks, the model advances the vision of transforming II from a passive infrastructure into an autonomic system that understands what it moves, why it moves, and whether it has logical permission to do so. The proposed architecture demonstrates how ontological rigor can ensure resilience, compliance, and semantic interoperability across heterogeneous domains, unifying domains (5G, satellite, SD-WAN) that would otherwise remain data silos. All results from this project will be made available at https://github.com/LJuliaoBraga/dtii/tree/main.

Future work will focus on: (i) simulation of the Digital Twin to evaluate alarms in Autonomous Systems (AS); (ii) experimental validation of spatio-temporal response mechanisms; (iii) integration with programmable infrastructures via ***NETCONF***; (iv) exploration of semantic interoperability across domains such as 5G, satellite, and SD-WAN; and (v) participation in the evaluation of the two proposed telemetry alternatives—YANG-Push and gNMI—while remaining open to those that may emerge in the near future; (vi) systematic documentation and evaluation of the ontology's competency questions and design rationale, clarifying how the conceptualization satisfies its intended modeling requirements; and (vii) empirical assessment of the completeness/efficiency trade-off between the current SROIQ-based reasoning and lightweight alternatives (OWL 2 RL/RDFox, OWL 2 QL/OBDA) under the proposed incremental, partition-based reasoning strategy.

These steps aim to evolve the design of the conceptual model to experimental deployment, culminating in its 2026 submission as a draft to the IRTF ***nmrg*** within the CGI-IETF initiative, in collaboration with the following Brazilian universities: **IFSPe** (PE), **ITA** (SP), **UFABC** (SP), **UFRGS** (RS), and **UNICAP** (PE).

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Statement on Generative AI

During the preparation of this work, the authors used Gemini and Copilot exclusively for grammar and spelling checks in Portuguese and English. Support was obtained for translations and relevant evaluations. All substantive content, analysis and conclusions were produced by the authors, who reviewed and edited the text as necessary and assume full responsibility for the publication's content.

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